The idea of particles, Copenhagen claims, is complementary to the idea of waves. The ideas are contradictory—photons cannot be both particles and waves—but both are necessary, in alternation, for describing this experiment. When you aren't measuring the position of a photon, it is a wave. Thus, photons can interfere with themselves as they pass through the double slit. But measuring the location of a photon forces it to behave as a particle: when the photon hits the screen behind the double slit, it must strike in only one spot. Similarly, putting photon detectors on each slit causes the photon to behave as a particle as it passes through the double slit: the detectors force each photon to pass through only one slit, and thus not interfere with themselves, when before they were free to behave as waves and pass through both slits. But asking where the photon was before the measurement is meaningless: waves have no singular location. The property measured was created by the measurement itself, and to ask about its location beforehand is mere sophistry. Any attempt to picture how this is possible, any attempt to give an account of how the quantum world behaves between measurements, is doomed to fail, because, as Bohr said, there is no quantum world.

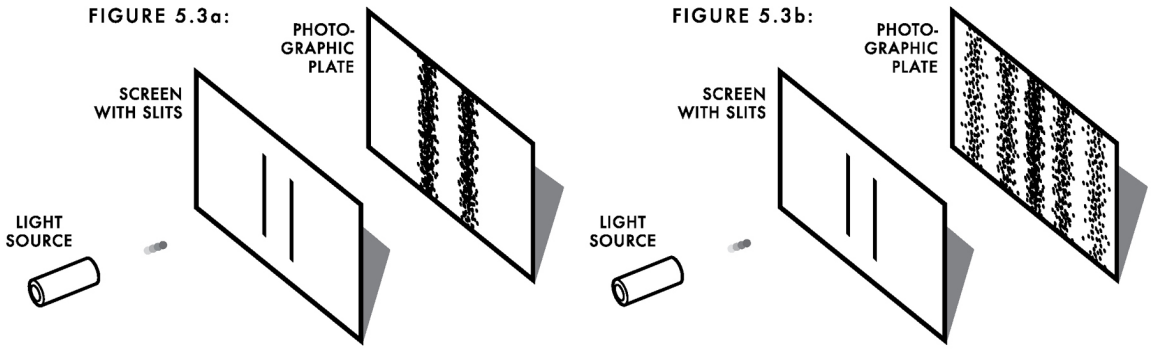


Figure 5.3. (a) We wouldn't expect individual photons passing one at a time through the double slit to produce an interference pattern. (b) Somehow, individual photons passing through the double slit do manage to interfere with themselves.

Bohm accounted for the strange results of the double-slit experiment by doing exactly what the Copenhagen interpretation said was impossible: he gave a detailed account of what happens in the quantum world whether or not anyone is looking. Photons, according to Bohm, are particles surfing on